

# Probabilistic Methods

## Lecture Notes

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June 5, 2026

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# Chapter 1

## Introduction

These notes introduce probabilistic methods as a way of reasoning about uncertainty. The course begins with sample spaces and events, then develops counting techniques, probability distributions, standard distribution functions, conditional probability, and finally descriptive statistics based on generated data.

The central aim is to connect formal probability with interpretation. A probability calculation should not be treated as a mechanical substitution into a formula. It should begin with a clear description of the random experiment, the elementary outcomes, the events or random variables of interest, and the assumptions that make a chosen model appropriate.

## Chapter 2

# Elements of Combinatorics

### 2.1 What combinatorics studies

Combinatorics is the mathematics of counting possibilities. It studies how many objects can be formed when a set of rules is imposed. These objects may be arrangements, lists, selections, strings, words, groups, schedules, routes, or distributions of identical items among different boxes.

The central question is simple:

How many objects of this type can be formed?

The difficulty is not usually arithmetic. The difficulty is recognizing the structure of the construction. Before using a formula, we must understand what is being built.

In this chapter we will not use the language of probability. We will also not start from abstract set theory. Instead, we will build a practical and precise language for counting. Later, this language will be useful in many places, but here it is studied on its own.

#### Counting as a construction process

A counting problem should be read as a recipe for constructing objects. For example, suppose we want to form a three-letter code from the letters

$$A, B, C, D.$$

Before counting, we must ask:

- Does the order of the letters matter?
- May a letter be used more than once?
- Do we use all available letters, or only some of them?
- Are some objects indistinguishable from each other?

Different answers lead to different counting schemes.

#### Theorem

**Multiplication principle.** If a construction is performed in several successive steps, and if there are  $m_1$  choices for the first step,  $m_2$  choices for the second step, and so on up to  $m_k$  choices for the  $k$ -th step, then the total number of possible constructions is

$$m_1 m_2 \cdots m_k.$$

The multiplication principle is the engine behind most formulas in this chapter. Factorials, powers, binomial coefficients, and stars-and-bars formulas all arise by applying this principle carefully and then correcting for possible overcounting.

## The six basic counting schemes

We will study six basic schemes. They are organized by three questions: whether order matters, whether repetition is allowed, and whether all objects are used or only a fixed number of positions is filled.

| scheme                          | order matters? | repetition allowed?            |
|---------------------------------|----------------|--------------------------------|
| permutations without repetition | yes            | no                             |
| permutations with repetition    | yes            | yes, through identical objects |
| variations without repetition   | yes            | no                             |
| variations with repetition      | yes            | yes                            |
| combinations without repetition | no             | no                             |
| combinations with repetition    | no             | yes                            |

The formulas are important, but the main goal is recognition. A formula used in the wrong situation gives a wrong answer even if the arithmetic is perfect.

### Remark

When solving a counting problem, do not start by asking which formula you remember. Start by asking what kind of object is being constructed.

## 2.2 Permutations without repetition

Combinatorics begins with arrangements. Imagine a row of empty places and a collection of distinct objects. We want to fill all places, using every object exactly once. In this situation, changing the order changes the result.

This is the most basic ordered counting scheme. It appears whenever the question is about arranging, ranking, scheduling, ordering, or placing all available distinct objects in a line.

### Definition

Let  $A$  be a finite set with  $n$  distinct elements. A permutation without repetition of the elements of  $A$  is an ordering of all elements of  $A$ , in which every element appears exactly once.

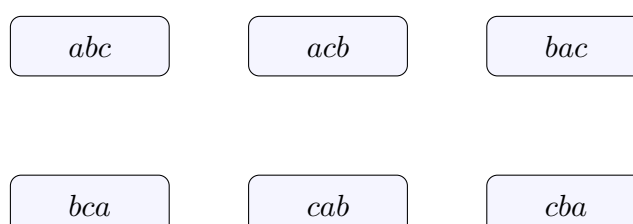
For example, if

$$A = \{a, b, c\},$$

then the permutations without repetition are

$$abc, \quad acb, \quad bac, \quad bca, \quad cab, \quad cba.$$

There are six of them.



### Theorem

The number of permutations without repetition of  $n$  distinct elements is

$$n! = n(n-1)(n-2) \cdots 2 \cdot 1.$$

The formula follows directly from the multiplication principle. There are  $n$  choices for the first place. Once the first object has been used, there are only  $n-1$  choices for the second place. After two objects have been used, there are  $n-2$  choices for the third place. Continuing this reasoning gives

$$n(n-1)(n-2) \cdots 2 \cdot 1 = n!.$$

The decreasing number of choices is the signature of “without repetition”. Each decision removes one object from later use.

### Example

Five different books are to be arranged on a shelf. Since all books are used, all books are distinct, and order matters, this is a permutation without repetition. The number of possible arrangements is

$$5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120.$$

### Example

Four students are to stand in a single line for a photograph. The students are distinct, every student stands in the line, and changing the order gives a different line. The number of possible lines is

$$4! = 24.$$

### Remark

Use permutations without repetition when all objects are distinct, all objects are used, and order matters. Typical words are: arrange, order, rank, line up, schedule all, place all.

## 2.3 Permutations with repetition

Permutations without repetition assume that all objects are distinct. In many counting problems this is not true. Some objects may look the same, carry the same label, or belong to the same type. Then exchanging two identical objects does not create a new visible arrangement.

The important idea is overcounting. If we temporarily pretend that all objects are distinct, we count too many arrangements. We then divide by the number of internal swaps that cannot be seen in the final result.

### Definition

A permutation with repetition is an ordering of a multiset, that is, a finite collection in which some elements may occur more than once.

For example, the letters in the word

$$AAB$$

are not all distinct. The two letters  $A$  are indistinguishable. The distinct arrangements are

$$AAB, \quad ABA, \quad BAA.$$

There are three, not six.

If all three positions were occupied by distinct objects, there would be  $3!$  arrangements. But the two identical  $A$ 's can be swapped without producing a new visible arrangement. Therefore we divide by  $2!$ .

### Theorem

Suppose a multiset has  $n$  elements in total, with  $n_1$  identical elements of one kind,  $n_2$  identical elements of another kind, and so on, where

$$n_1 + n_2 + \cdots + n_k = n.$$

The number of distinct permutations is

$$\frac{n!}{n_1!n_2!\cdots n_k!}.$$

The numerator  $n!$  counts arrangements as if every object were distinct. The denominator removes invisible rearrangements inside each group of identical objects. If one kind appears  $n_i$  times, then those  $n_i$  identical copies can be internally rearranged in  $n_i!$  ways without changing the final visible arrangement.

### Example

How many distinct arrangements are there of the letters in the word LEVEL?

There are five letters. The letter  $L$  appears twice, the letter  $E$  appears twice, and the letter  $V$  appears once. Hence

$$\frac{5!}{2!2!1!} = \frac{120}{4} = 30.$$

### Example

How many distinct strings can be formed from the letters of AABBC?

There are five positions. The letter  $A$  appears twice,  $B$  appears twice, and  $C$  appears once. Therefore the number is

$$\frac{5!}{2!2!1!} = 30.$$

### Remark

Use permutations with repetition when all positions are filled, order matters, and some objects are indistinguishable. Typical examples are arrangements of letters in a word with repeated letters or arrangements of objects classified by type rather than identity.

## 2.4 Variations without repetition

A permutation uses all available objects. A variation uses only some of them. The order still matters, but we stop after filling  $k$  positions.

This situation appears when we form a short ordered list from a larger collection: a podium with first, second, and third place; a code made of different symbols; a sequence of selected objects; or an ordered assignment of distinct objects to distinct positions.

### Definition

Let  $A$  be a finite set with  $n$  distinct elements, and let  $k \leq n$ . A variation without repetition of length  $k$  is an ordered selection of  $k$  distinct elements from  $A$ .

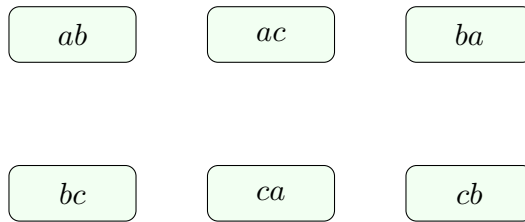
For example, from

$$A = \{a, b, c\}$$

the variations without repetition of length 2 are

$$ab, \quad ac, \quad ba, \quad bc, \quad ca, \quad cb.$$

The selections  $ab$  and  $ba$  are different, because order matters.



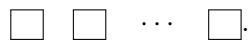
### Theorem

The number of variations without repetition of length  $k$  chosen from  $n$  distinct elements is

$$V(n, k) = n(n-1)(n-2) \cdots (n-k+1) = \frac{n!}{(n-k)!}.$$

There are  $n$  choices for the first position,  $n-1$  choices for the second position, and so on. Since no element may be repeated, the number of choices decreases after each position is filled. We stop after  $k$  positions, so the product has only  $k$  factors.

A useful mental picture is a row of  $k$  boxes:



The first box has  $n$  possible fillings, the second has  $n-1$ , and the process continues until the  $k$ -th box.

### Example

A code consists of three different letters chosen from the six letters

$$A, B, C, D, E, F.$$

Order matters and no letter may repeat. The number of possible codes is

$$V(6, 3) = 6 \cdot 5 \cdot 4 = 120.$$

### Example

Eight runners compete for three medals: gold, silver, and bronze. The order matters because the medals are different, and no runner can receive more than one medal. Thus the number of possible medal outcomes is

$$V(8, 3) = 8 \cdot 7 \cdot 6 = 336.$$



### Remark

Use variations without repetition when order matters, only  $k$  positions are filled, and no object may be used more than once. This is the ordered-selection scheme.

## 2.5 Variations with repetition

In some ordered selections, the same object may be used more than once. Then every position can be filled independently from the same collection of choices. The number of available choices does not decrease after a position is filled.

This is the natural scheme for strings, passwords, codes, sequences of symbols, repeated choices, and any construction in which the same label may appear several times.

### Definition

Let  $A$  be a finite set with  $n$  distinct elements. A variation with repetition of length  $k$  is an ordered  $k$ -tuple of elements of  $A$ , where elements may repeat.

For example, if

$$A = \{a, b\},$$

then the variations with repetition of length 3 are

$$aaa, \quad aab, \quad aba, \quad abb, \quad baa, \quad bab, \quad bba, \quad bbb.$$

There are 8 of them. The word  $aba$  is different from  $baa$ , because order matters. The word  $aaa$  is allowed, because repetition is allowed.

### Theorem

The number of variations with repetition of length  $k$  chosen from  $n$  distinct elements is

$$n^k.$$

There are  $n$  choices for the first position. Because repetition is allowed, there are again  $n$  choices for the second position, again  $n$  choices for the third position, and so on. Therefore the multiplication principle gives

$$\underbrace{n \cdot n \cdots n}_{k \text{ factors}} = n^k.$$

The constant number of choices is the signature of repetition. Each position is filled from the full original set.

### Example

A three-symbol string is formed from the alphabet

$$\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}.$$

Symbols may repeat. There are

$$10^3 = 1000$$

such strings.

### Example

A lock code has four positions. Each position may contain one of the digits  $0, 1, \dots, 9$ , and digits may repeat. The number of possible codes is

$$10^4 = 10000.$$

### Remark

Use variations with repetition when order matters, there are  $k$  positions, and every position can be filled independently from the same set of  $n$  objects. Typical words are: string, password, code, sequence, ordered tuple.

## 2.6 Combinations without repetition

In combinations, order does not matter. We are interested only in which objects are selected, not in the order in which they are written.

This is a different way of thinking from permutations and variations. In an ordered list,  $ab$  and  $ba$  are different. In an unordered selection, they represent the same choice. The objects are collected into a group, not placed into positions.

### Definition

Let  $A$  be a finite set with  $n$  distinct elements, and let  $k \leq n$ . A combination without repetition of size  $k$  is a  $k$ -element subset of  $A$ .

For example, if

$$A = \{a, b, c\},$$

then the combinations without repetition of size 2 are

$$\{a, b\}, \quad \{a, c\}, \quad \{b, c\}.$$

The expressions  $\{a, b\}$  and  $\{b, a\}$  describe the same set, so they are not counted separately.

$$\begin{array}{ccc} \begin{array}{c} \textcircled{a} \quad \textcircled{b} \\ \{a, b\} \end{array} & \begin{array}{c} \textcircled{a} \quad \textcircled{c} \\ \{a, c\} \end{array} & \begin{array}{c} \textcircled{b} \quad \textcircled{c} \\ \{b, c\} \end{array} \end{array}$$

### Theorem

The number of combinations without repetition of size  $k$  chosen from  $n$  distinct elements is

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

To see why, first count ordered selections without repetition. There are

$$V(n, k) = \frac{n!}{(n-k)!}$$

of them. But each unordered group of  $k$  elements can be ordered in  $k!$  ways. Therefore each combination has been counted  $k!$  times. Dividing by  $k!$  gives

$$\frac{1}{k!} \cdot \frac{n!}{(n-k)!} = \frac{n!}{k!(n-k)!}.$$

This division by  $k!$  is not a trick. It removes the internal orderings of the same selected group.

### Example

From six distinct books, choose three books to take on a trip. The order of choosing does not matter. The number of possible selections is

$$\binom{6}{3} = \frac{6!}{3!3!} = 20.$$

### Example

From the set  $\{1, 2, 3, 4, 5\}$ , choose two numbers. The possible groups are not ordered, so  $\{1, 4\}$  and  $\{4, 1\}$  are the same group. The number of choices is

$$\binom{5}{2} = 10.$$

### Remark

Use combinations without repetition when order does not matter and no object may be selected more than once. Typical words are: choose, select, form a group, form a committee, pick a subset.

## 2.7 Combinations with repetition

The last basic scheme combines two ideas: order does not matter, but repetition is allowed. We select a group of objects, and the same kind of object may be selected more than once.

This scheme is less obvious than the previous ones, because the objects we are counting are not naturally written as ordered strings. A selection such as

$$\{a, a, c\}$$

contains two copies of  $a$ , no copies of  $b$ , and one copy of  $c$ . The order in which we write these symbols is irrelevant.

### Definition

Let  $A$  be a finite set with  $n$  distinct elements. A combination with repetition of size  $k$  is a selection of  $k$  elements from  $A$ , where elements may repeat and order does not matter.

For example, if

$$A = \{a, b, c\}$$

and we select 2 elements with repetition allowed, then the possible selections are

$$\{a, a\}, \quad \{a, b\}, \quad \{a, c\}, \quad \{b, b\}, \quad \{b, c\}, \quad \{c, c\}.$$

The selections  $\{a, b\}$  and  $\{b, a\}$  are the same, because order does not matter.

## Counting by multiplicities

A useful way to count such selections is to record only how many times each element is chosen. If the elements are

$$a_1, a_2, \dots, a_n,$$

then a selection of size  $k$  corresponds to a solution of

$$x_1 + x_2 + \dots + x_n = k,$$

where each  $x_i$  is a non-negative integer. The number  $x_i$  tells how many times  $a_i$  is selected.

For example, if  $A = \{a, b, c\}$ , then the selection  $\{a, a, c\}$  corresponds to

$$x_a = 2, \quad x_b = 0, \quad x_c = 1.$$

### Theorem

The number of combinations with repetition of size  $k$  chosen from  $n$  distinct elements is

$$\binom{n+k-1}{k} = \binom{n+k-1}{n-1}.$$

One way to understand the formula is the stars and bars method. Imagine  $k$  identical stars and  $n - 1$  bars separating  $n$  boxes. The stars in the first box count how many times the first element is chosen, the stars in the second box count how many times the second element is chosen, and so on.

For example, with  $n = 3$  and  $k = 5$ , the pattern

$$\star\star \mid \star \mid \star\star$$

represents

$$x_1 = 2, \quad x_2 = 1, \quad x_3 = 2.$$

Altogether there are  $k + n - 1$  symbols:  $k$  stars and  $n - 1$  bars. Choosing the positions of the stars determines the bars, and choosing the positions of the bars determines the stars. Therefore the number of patterns is

$$\binom{k+n-1}{k} = \binom{k+n-1}{n-1}.$$

### Example

A bag is to contain 4 candies chosen from 3 flavours: vanilla, chocolate, and strawberry. Repetition is allowed, and order does not matter. The number of possible bags is

$$\binom{3+4-1}{4} = \binom{6}{4} = 15.$$

### Example

How many non-negative integer solutions are there to

$$x_1 + x_2 + x_3 + x_4 = 7?$$

This is the same as distributing 7 identical stars among 4 boxes. Hence the number of

solutions is

$$\binom{7+4-1}{7} = \binom{10}{7} = 120.$$

### Remark

Use combinations with repetition when order does not matter and each type of object may be selected more than once. Typical words are: choose with repetition, distribute identical objects, non-negative integer solutions, multiset.

## 2.8 How to recognize the counting scheme

The six formulas in this chapter are easy to confuse if they are memorized as isolated expressions. A safer method is to classify the construction before choosing a formula.

Start with the following questions.

1. Are we arranging all objects, or choosing only  $k$  of them?
2. Does order matter?
3. Can an object be used more than once?
4. Are some objects indistinguishable?

These questions usually determine the correct scheme.

| Situation                    | Order           | Repetition        | Formula                       |
|------------------------------|-----------------|-------------------|-------------------------------|
| arrange all distinct objects | matters         | no                | $n!$                          |
| arrange a multiset           | matters         | identical objects | $\frac{n!}{n_1! \cdots n_r!}$ |
| choose ordered $k$ objects   | matters         | no                | $\frac{(n-k)!}{n^k}$          |
| form ordered $k$ -tuple      | matters         | yes               | $n^k$                         |
| choose $k$ -element subset   | does not matter | no                | $\binom{n}{k}$                |
| choose multiset of size $k$  | does not matter | yes               | $\binom{n+k-1}{k}$            |

### Remark

A common mistake is to decide too quickly whether order matters. Ask whether swapping two selected objects creates a genuinely different object. If yes, order matters. If no, order does not matter.

### Example

Suppose we choose three people from ten to form a committee. The word committee suggests that the selected people form a group, not an ordered list. Thus order does not matter, and no person can be chosen twice. The correct scheme is combinations without repetition:

$$\binom{10}{3}.$$

### Example

Suppose we choose a president, a secretary, and a treasurer from ten people. Now the roles are different, so assigning Alice as president and Bob as secretary is different from assigning Bob as president and Alice as secretary. Order, or more precisely position, matters. No person can hold two roles. The correct scheme is variations without repetition:

$$\frac{10!}{(10 - 3)!} = 10 \cdot 9 \cdot 8.$$

The mathematical formulas are compact, but the real skill is translating a verbal description into a construction. Once the construction is understood, the formula usually becomes natural.

## Chapter 3

# Events and Probability (Sample Spaces)

This chapter introduces the fundamental concepts of probability theory, including sample spaces, events, and the calculation of probabilities.

## Chapter 4

# Probability Distributions (Selected Models)

This chapter introduces several important probability distributions that are used to model various random phenomena.



## Chapter 5

# Discovering Formalism

This chapter focuses on the conceptual development of probability theory, moving from intuitive ideas to a more formal mathematical framework.

## Chapter 6

# PMF, PDF, and CDF

This chapter provides a technical overview of the standard representations of probability distributions: the Probability Mass Function (PMF), the Probability Density Function (PDF), and the Cumulative Distribution Function (CDF).

## Chapter 7

# Algebra of Events and Conditional Probability

This chapter explores the algebra of events and the concept of conditional probability, which are fundamental to understanding how probabilities are updated with new information.

## Chapter 8

# Descriptive Statistics from Generated Data

This chapter shifts the perspective from theoretical probability models to the analysis of data using descriptive statistics.